Malhar Gudekar

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Education

University of Illinois Urbana Champaign August 2024 - May 2026
Master of Science, Information Management GPA: 3.8/4
• Coursework: Web Programming, Machine Learning & Cloud, Information Management, Information Consulting, Methods of Data Science

University of Mumbai June 2019 - May 2023
Bachelor of Engineering, Computer Science GPA: 9.25/10
• Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, NLP

Skills

- **Languages & DSA:** Python, SQL, Data Structures & Algorithms (Arrays, Hashing, Trees, Graphs, Dynamic Programming), LeetCode
- **Backend & Systems:** FastAPI, REST APIs, Microservices, Distributed Systems, ETL Pipelines, Kafka, PySpark
- **Databases, Cloud & Tools:** PostgreSQL, Neo4j, AWS (S3, EC2, EKS), GCP, Docker, Git

Work Experience

University of Illinois Urbana Champaign May 2025 - Present
Research Assistant - iSchool Champaign, USA
- Owned the **end-to-end design, development, and integration** of cross-platform system ingesting wearable data for 100+ users, enabling scalable, continuous data collection and downstream analytics
- Designed and optimized **PostgreSQL schema, indexing strategies, and query execution plans** for high-volume datasets, improving query performance by 38% under production workloads
- Built robust, production-ready **Python** pipeline with retrieval and context summarization, improving multi-turn response accuracy by **43%**
- Developed **backend workflows** for real-time, fault-tolerant processing of longitudinal health data with focus on reliability and scalability

Research Assistant - CHI AI in Infodemic Management January 2025 - May 2025
- Designed and deployed highly **scalable NLP pipelines** in **Python** to process high-volume datasets, reducing processing latency by **41%**
- Built distributed data ingestion pipelines using **Kafka** and **Neo4j** Streams for real-time, high-throughput processing and pattern detection
- Designed and developed **production-grade REST APIs using FastAPI**, implementing validation, error handling, and monitoring, and built automated deployment workflows to ensure **high reliability, reproducibility, and scalable integration** across distributed systems
- Enabled large-scale data processing across social media streams using **distributed system architecture** principles

Business Intelligence Group August 2025 - December 2025
Technical Consultant Champaign, USA
- Built and deployed scalable, microservices-based Retrieval-Augmented Generation (**RAG**) system using **FastAPI**, **PostgreSQL**, and vector embeddings for production-grade applications
- Designed efficient, fault-tolerant **ETL** pipelines to process and index large document datasets for low-latency, high-throughput retrieval systems
- Improved system performance by **64%** through optimized indexing strategies and pipeline efficiency improvements in production environments
- Developed backend services enabling real-time, high-performance querying of unstructured data across distributed system architectures

Swift Mobil Software Solutions Provider July 2023 - October 2023
Data Analyst Mumbai, IN
- Built interactive **Power BI** dashboards for **logistics** and **mobility KPIs**, reducing reporting time by **40%** and improving visibility into operational performance across transportation services and decision-making insights
- Processed and optimized large-scale operational datasets using **Python** and **PySpark** to identify failure patterns, trends, and performance issues in distributed systems environments
- Developed **SQL** and **graph queries** to map dependencies, analyze operations, detect data quality issues, speeding root-cause analysis by **26%**

Pscope Technologies Pvt. Ltd. January 2023 - June 2023
Data Analyst Mumbai, IN
- Developed and maintained **20+ Power BI** dashboards across Sales, Finance, and Operations for enterprise clients, leveraging **SQL** for data extraction and **DAX** for advanced calculations, enabling KPI tracking and data-driven decision-making
- Automated recurring **Excel**-based reporting workflows using **VBA**, implementing data validation, transformation, and scheduling logic, reducing report preparation time by **42%** and improving data consistency across weekly reporting cycles

Stu/Dio at Illinois August 2025 - Present
Project Manager Champaign, USA
- Led cross-functional team of **6** engineers and designers to deliver product updates for DeepCover game, leveraging user feedback, telemetry data, and performance insights to drive data-driven feature prioritization and improvements
- Tracked **sprint** progress, delivery metrics, and **QA issues** through **JIRA**-based workflows, managing backlog grooming, sprint planning, and issue resolution to improve cross-team visibility and ensure on-time milestone delivery

Projects

Scalable ML System for Customer Risk Prediction March 2025
UIUC Hackathon
- Built an **end-to-end machine learning system** using Python, Pandas, and LightGBM to forecast customer spending and assess credit risk from large-scale **time-series** transaction data, including data preprocessing, feature engineering, model training, and evaluation
- Designed and implemented **scalable ETL and feature engineering pipelines**, capturing behavioral trends and temporal patterns such as rolling averages and spending frequency to improve model performance
- Developed a **decision framework for credit line adjustments**, incorporating model outputs and risk thresholds to improve prediction precision by 20% and enable risk-aware classification of customers